

Adaptive DBSCAN LiDAR Point Cloud Clustering For Autonomous Driving Applications

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Abstract— *LiDAR point cloud clustering is an essential part of a wide range of applications such as object detection, object recognition, and localization. In this paper, we focus on Density Based Spatial Clustering of Applications with Noise (DBSCAN) as a very promising and efficient algorithm to cluster LiDAR point cloud. However, it requires two tuning parameters that are manually specified based on prior knowledge of the domain. Furthermore, the selected tuning parameters cannot be modified once the clustering process begins. Environments in domains like robotics and autonomous driving are very dynamic, and hence, automated and adaptive selection of these parameters is preferred. In this work, we propose to estimate DBSCAN tuning parameters automatically based on a field of view division scheme, and points' distances and densities within the division scheme. Furthermore, we adaptively estimate the tuning parameters for every point cloud frame separately. We evaluated the proposed method using the KITTI dataset, and the preliminary results show that the proposed method can achieve comparable results to the original DBSCAN without the need for manually selecting the parameters.*

Keywords— *LiDAR, Density-Based Clustering, DBSCAN, Object detection*

I. INTRODUCTION

Light Detection and Ranging (LiDAR) sensors are an essential part of most autonomous solutions like robotics and self-driving cars. LiDAR generates dense unordered 3-Dimensional (3D) information which is very difficult to work with directly, therefore different preprocessing operations are usually conducted to filter out noise information and extract useful information about the remaining point cloud [1-7]. Clustering LiDAR data is very beneficial to many applications, especially those edge-based real-time applications like object detection and classification. Density-based clustering algorithms group objects based on density, wherein objects are within high-density regions and separated by low-density regions – usually noise. Such a characteristic makes them a good candidate for LiDAR point cloud clustering [6-13]. Density-Based Spatial Clustering of Applications with Noise (DBSCAN) [14] cluster data based on the density, and it does not require a prior assignment for clusters number which is a typical scenario in dynamic environments like robotics and self-driving environments. DBSCAN uses two parameters to define density level and extract clusters information: ϵ (epsilon), which is the point neighborhood search distance, and minPts , which is the

minimum number of neighboring points needed to form a cluster from a given point. These tuning parameters are selected based on the dataset and then utilized by the clustering algorithm.

DBSCAN defines three types of points: core, border, and noise. A core point in the dataset has at least minPts of points within the ϵ distance. A border point in the dataset is within another core point, and has fewer neighboring points than minPts . A noise point is outside any core point range and also has fewer neighboring points than minPts . DBSCAN also uses three main relations to grow clusters: directly density-reachable, the density-reachable, and the density connected relations. The directly density reachable relation can be defined as follow: for any two points m and n , a point m is directly density reachable from point n if m belongs to the n points neighborhood and n is a core point. However, in the density reachable relation, a point m is density reachable from point n if there is a chain of points where each consecutive points are directly density reachable. Finally, in the density connected relation, point m is density connected to point n if

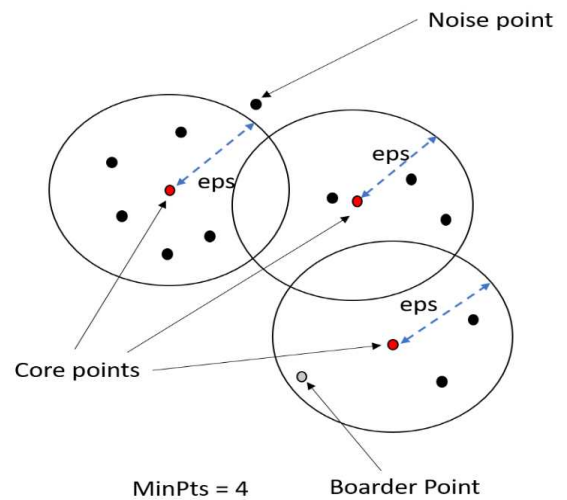


Figure 1. DBSCAN point types: red points are core points, gray point is a border point and black point outside the circles are noise point [14]

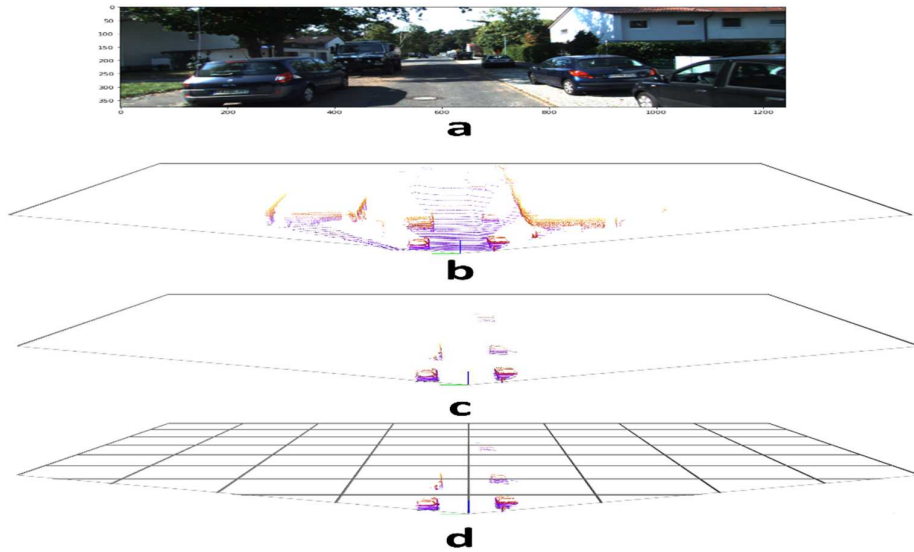


Figure 2. a) Reference frame from camera b) LiDAR point cloud before background removal c) LiDAR point cloud after background removal d) LiDAR foreground point cloud with the proposed division scheme

there is another point p which is density reachable from both m and n . Figure 1 depict the main point types in DBSCAN.

In this work, we propose to estimate DBSCAN parameters ϵ and $\minPts$ for each point in each point cloud frame separately, thus each point will have its own ϵ and $\minPts$. We divide the field of view (FOV) into different regions and further divide each region into multiple cells, then we calculate different interesting quantities to estimate the tuning parameters. Our method will result in more accurate clusters, especially for objects far from the LiDAR.

II. RELATED WORK

DBSCAN is a very popular density-based clustering algorithm, and it has been utilized in different domains and applications [15-20]. Although DBSCAN is a very popular density-based clustering algorithm, limited work has been proposed to estimate the tuning parameters in the autonomous and robotics contexts. However, the following works are the most relevant to our work. Reference [5] proposed an estimation approach that is based on the average K-NN. It first normalizes the data and then it builds a spatial index, then using both the spatial index and different distances calculations it estimates the clustering parameters, then the clustering process takes place using the different estimated ϵ . Reference [21] proposed Adaptive Searching-DBSCAN which starts with initial values and then revises it in each clustering iteration. The authors suggested equations to estimate both the initial values and the subsequent values. The proposed equations take into consideration different parameters like expected target width and expected number of points per location. However, none of the equations took into consideration the location of the LiDAR or the relative distance of the points to the LiDAR. Reference [22] uses the infrastructure LiDAR's to detect and track pedestrians at intersections, to improve DBSCAN and automatically select the tuning parameters, they took into consideration the properties of the installed LiDAR and they introduced two equations to find both the minimum horizontal and minimum vertical distance between LiDAR points, also three zones division scheme is introduced to change the values of the

tuning parameters based on the object's locations. In reference [23] the authors proposed an unsupervised clustering technique that consists of different stages like data normalization, spatial index, parameters estimation, and clustering. Parameter's estimation is done using the mean distance of the points, the process selects subgroup of the points and then finds the neighbors by KNN search then find the distance between the selected points and their points and finally calculate the mean and use it in the parameter estimation.

III. THE PROPOSED METHOD

To meet our goals of estimating DBSCAN clustering parameters automatically and adaptivity based on the point cloud structure; we propose a field of view division technique and empirical relations that accurately estimate both ϵ and $\minPts$ for each point individually. Before starting the parameter estimation, we filter out the background points by utilizing a simple but efficient deep neural network (DNN) similar to the network of [24]. This network has a decoder/encoder structure which learns unique features that distinguish between foreground and background points. The resulting features vector is used to classify each point into foreground or background. An example of LiDAR point cloud before and after background filtration is depicted in Figure 2-b and Figure 2-c respectively. Background filtration process will reduce the point cloud size and allow for a faster clustering process. The proposed scheme is based on dividing the FOV into equal-sized regions and further divide each region into equal-sized cells. An illustrative example is shown Figure 2-d. This division process will allow for local information about the LiDAR point cloud to be computed, like density of the points within each cell.

A. Eps estimation

As defined in DBSCAN [14], ϵ is the cluster radius, and it is used to check if a point is a core point, border point, or noise point. Points near the LiDAR tend to be dense and very close to each other, while points far from the LiDAR tend to be sparse and thus not close to each other. Therefore, ϵ is better

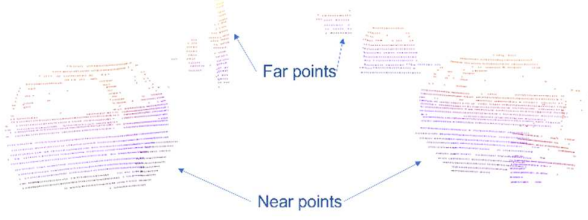


Figure 3. A zoomed LiDAR point cloud view shows near and far points

to be small for near points and get larger while distance from the LiDAR increase. Figure 3 shows a zoomed view of LiDAR point cloud. As we can see from the figure, near points are dense and very close to each other while far points are sparse and far from each other. From a density perspective, higher density implies more adjacent points and thus smaller ϵ is better while lower density implies fewer adjacent points and thus larger ϵ is better. Mathematically these relations can be represented as in relation 1 below:

$$\epsilon \propto d \text{ and } \epsilon \propto \frac{1}{\text{density}} \quad (1)$$

where d is the Euclidean distance between the point and the LiDAR, and the density is the total number of points in cell per m^2 .

B. Minpts estimation

DBSCAN [14] defines minPts as the minimum number of neighboring points needed to form a valid cluster from a given point. minPts for dense objects can be small however if an object is sparse minPts should be larger. Moreover, as ϵ increases, wider area will be scanned and minPts will consequently increase. Mathematically these relations can be represented as in relation 2 below:

$$\text{minPts} \propto \epsilon \text{ and } \text{minPts} \propto \frac{d}{\text{density}} \quad (2)$$

Figure 4 shows the proposed FOV division scheme and parameters definitions.

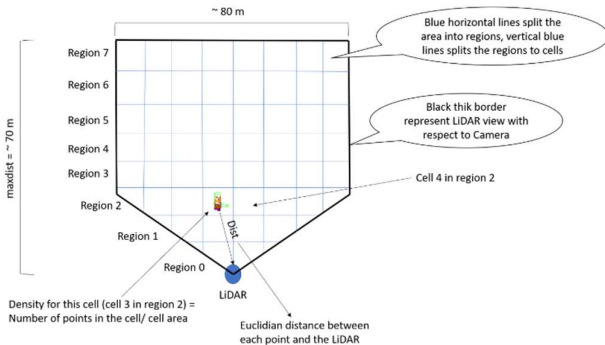


Figure 4. Proposed FOV division and parameters definitions

IV. PRELIMINARY RESULTS

The performance of our proposed approach is evaluated on the KITTI dataset [25], KITTI provides ground truth information about objects in each frame in the form of 3-D and 2-D bounding boxes. KITTI was collected in Karlsruhe, Germany using a VW station wagon equipped with different sensors like Cameras, LiDAR, and high-precision GPS/IMU. Therefore, we apply our clustering method and then create bounding boxes and compare it to the available ground truth bounding boxes. Figure 5 provides a visual performance comparison between the original DBSCAN and the proposed adaptive method illustrated in 2D domain for better visualization, Figure 5-a provides the reference 2D camera image, Figure 5-b shows the performance of the original DBSCAN, and Figure 5-c shows the performance of the proposed method. Our adaptive method correctly distinguishes between the adjacent cars indicated by the gray arrow while the original DBSCAN fails and merges them into one single object. Figure 6 shows a comparison between the original DBSCAN and the proposed method in terms of correctly identified vs the incorrectly identified objects. The figure clearly shows that our proposed method achieves comparable performance to the original DBSCAN, but without the need to conduct several trials to find and manually select the tuning parameters. It is worth to mention that for the original DBSCAN, we conducted twenty-five experiments to choose the best tuning parameters. On the other hand, our proposed approach estimated the tuning parameters automatically and adaptively from the first time. This is very useful in real-time applications like robotics and autonomous vehicles wherein the environment is dynamic and fixed parameters are inefficient.

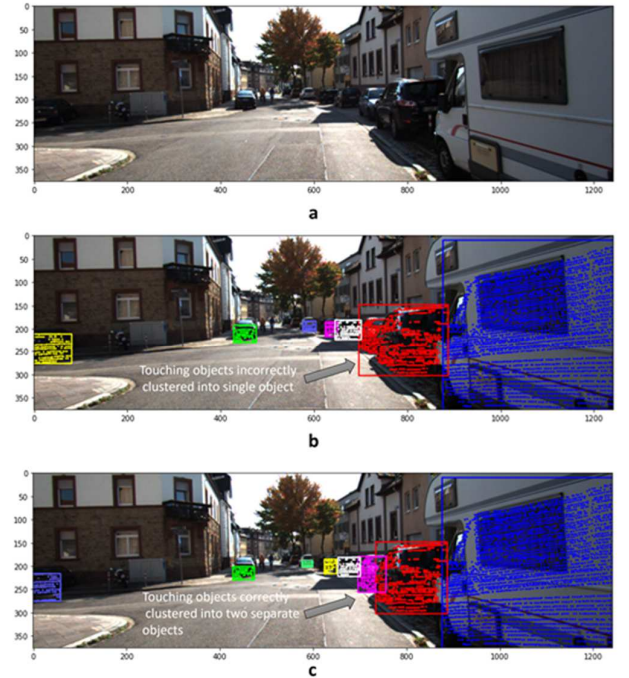


Figure 5. Visual results for a selected frame. a) reference frame, b) LiDAR point cloud clustered and projected into 2D image using the non-adaptive DBSCAN c) LiDAR point cloud clustered and projected into 2D image using the proposed adaptive DBSCAN method

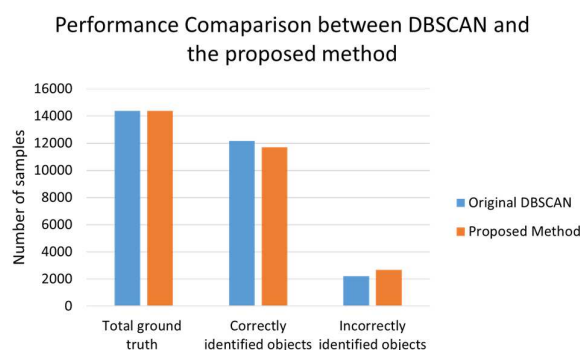


Figure 6. Performance of the proposed method and the original DBSCAN in terms of correctly identified objects and incorrectly identified objects

V. CONCLUDING REMARKS

In this paper, we proposed a method to automatically estimate the tuning parameters of DBSCAN for robotics and autonomous driving applications. In this method, the two tuning parameters ϵ and minPts are automatically estimated based on a field of view division scheme and empirically developed relations. The proposed relations relate the tuning parameters to the density of the points within cells and to the Euclidean distance between the points and the LiDAR. The proposed approach achieved comparable performance to the original DBSCAN while automating the process of selecting the tuning parameters and making them dynamic and unique for each point within the point cloud. Work will follow to fine-tune the division scheme and the proposed relationships to increase robustness and accuracy.

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